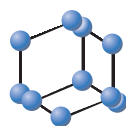
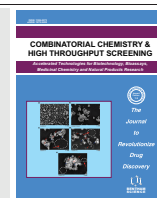


## RESEARCH ARTICLE

BENTHAM  
SCIENCE

# Network Pharmacology and Experimental Validation of Qingwen Baidu Decoction Therapeutic Potential in COVID-19-related Lung Injury



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**Abstract: Background and Purpose:** Coronavirus disease 2019 (COVID-19) is a life-threatening disease worldwide due to its high infection and serious outcomes resulting from acute lung injury. Qingwen Baidu decoction (QBD), a well-known herbal prescription, has shown significant efficacy in patients with Coronavirus disease 2019. Hence, this study aims to uncover the molecular mechanism of QBD in treating COVID-19-related lung injury.

**Methods:** Traditional Chinese Medicine Systems Pharmacology database (TCMSP), DrugBanks database, and Chinese Knowledge Infrastructure Project (CNKI) were used to retrieve the active ingredients of QBD. Drug and disease targets were collected using UniProt and Online Mendelian Inheritance in Man databases (OMIM). The core targets of QBD for pneumonia were analyzed by the Protein-Protein Interaction Network (PPI), Gene Ontology (GO), and Kyoto Encyclopedia of Genes and Genomes (KEGG) to reveal the underlying molecular mechanisms. The analysis of key targets using molecular docking and animal experiments was also validated.

**Results:** A compound-direct-acting target network mainly containing 171 compounds and 110 corresponding direct targets was constructed. The key targets included STAT3, c-JUN, TNF- $\alpha$ , MAPK3, MAPK1, FOS, PPARG, MAPK8, IFNG, NF $\kappa$ B1, *etc.* Moreover, 117 signaling pathways mainly involved in cytokine storm, inflammatory response, immune stress, oxidative stress and glucose metabolism were found by KEGG. The molecular docking results showed that the quercetin, alanine, and kaempferol in QBD demonstrated the strongest affinity to STAT3, c-JUN, and TNF- $\alpha$ . Experimental results displayed that QBD could effectively reduce the pathological damage to lung tissue by LPS and significantly alleviate the expression levels of the three key targets, thus playing a potential therapeutic role in COVID-19.

**Conclusion:** QBD might be a promising therapeutic agent for COVID-19 via ameliorating STAT3-related signals.

**Keywords:** Qingwen Baidu. network pharmacology, molecular docking, decoction, COVID-19, acute lung injury, WHO.

## 1. INTRODUCTION

Coronavirus disease 2019 (COVID-19) is a disease caused by a new coronavirus (nCoV), namely severe acute respiratory syndrome coronavirus type 2 “SARS-CoV-2” by the World Health Organization (WHO) [1]. Humans are susceptible to this highly pathogenic coronavirus, and patients may develop fever, malaise, dry cough, shortness of breath, upper respiratory tract congestion, and other symptoms in the early stages of infection, which may be accompanied by

muscle aches and sore throat. With the development of the disease, patients may develop critical conditions, such as systemic inflammatory response and respiratory distress at a later stage, and further develop acute respiratory distress syndrome, bleeding and coagulation dysfunction, and multi-organ failure, thus leading to death [2]. According to the WHO, from the outbreak of the epidemic in December, 2019, to July 1<sup>st</sup>, 2021 (as of 18:28 EST), SARS-CoV-2 has spread in more than 200 countries and regions, with a total of 181,930,736 confirmed cases reported worldwide, including 39,458,832 deaths; China reported 118,573 confirmed cases, including 5,495 deaths; the United States reported 333,7083 confirmed cases, including 599,089 deaths. It is clear that COVID-19 is highly prevalent. According to the

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COVID-19 treatment guidelines released in 2021 at home and abroad [3, 4], the current clinical treatment of neonconjunctivitis and its complications is often done by drugs, such as raltegravir, IL-6 inhibitors, interferon alpha, and dexamethasone to suppress the virus, excessive immunity, and inflammation [5], but so far there is still no specific drug for the treatment of COVID-19. SARS-CoV-2 may continue to affect human life for a long time in the future, so research on a vaccine to prevent COVID-19 is urgent. At present, all available vaccine technologies are being used worldwide to urgently develop vaccines, and nearly 200 vaccines are under development, of which nearly 100 COVID-19 vaccines are in the clinical research stage, and more than 10 vaccines have been used for population vaccination [6, 7].

In addition to COVID-19, multiple pandemic diseases caused by coronaviruses have emerged worldwide in the past two decades. The outbreaks of SARS-CoV (severe acute respiratory syndrome coronavirus), MERS-CoV (Middle East respiratory syndrome coronavirus), and coronary pneumonia in 2002 and 2012 caused even higher human mortality than COVID-19, causing a great loss to the productive life of human society [8-11]. Therefore, finding effective drugs, methods, and technologies for the prevention and treatment of pneumonia is a common struggle for medical practitioners worldwide.

COVID-19 belongs to the category of "epidemic disease" in Chinese medicine, and Chinese medicine has been handed down for thousands of years, with rich clinical experience and a complete theoretical system for fighting against the epidemic [12]. Chinese medicine has played a unique and important role in the prevention and treatment of COVID-19 in China [13]. In China, 92.41% of diagnosed cases have TCM involved in rescue and treatment, with an overall efficiency of 90% [14]. TCM interventions can shorten the healing time of patients, reduce the conversion rate of severe disease, and reduce the cost of treatment, so TCM has a role to play in the battle against the epidemic in China that cannot be ignored [15, 16]. Among them, the Chinese herbal medicine compound Qingwen Baidu decoction (QBD) (composed of Gypsum fibrosum, Radix Rehmanniae, Bubali Cornu, Radix Scutellariae, Rhizoma Coptidis, Radix Platycodi, Rhizoma Anemarrhenae, raw Fructus Gardeniae, Radix Paeoniae Rubra, Cortex Moutan, Radix Scrophulariae, Fructus Forsythiae, Herba Lophatheri, and Radix Glycyrrhizae) from the Qing Dynasty doctor Yu Lin's monograph on warm disease, "Epidemic Rash Yide", has the effect of expelling the pathogens from the lungs, detoxifying and cooling the blood, relieving fever, exerting anti-inflammatory and anti-viral effects, regulating immunity, *etc.* It is effective in the treatment of COVID-19 mild, common and heavy patients. and heavy patients with significant efficacy [17]. Therefore, this study used network pharmacology and molecular docking techniques to construct a complex network of "component-target-disease-pathway" to predict potential targets and related mechanisms. To validate the prediction results and increase the credibility of the study, we conducted anti-acute pneumonia experiments in C57BL/C mice to provide a basis for the treatment of COVID-19 and the prevention of future pneumonia diseases.

## 2. MATERIALS AND METHODS

### 2.1. Screening of Active Ingredients and Targets in QBD

Using the Traditional Chinese Medicine Systematic Pharmacology Database and Analysis Platform TCMSP (<http://tcmspw.com/tcmsp.php>), the 14 Chinese herbal medicines in Qingwen Baidu decoction were searched for Gypsum Fibrosum (Shigao), Radix Rehmanniae (Dihuang), Bubali Cornu (Shui Niu Jiao), Radix Scutellariae (Huang Qin), Rhizoma Coptidis (Huang Lian), Radix Platycodi (Jie Geng), Rhizoma Anemarrhenae (Zhi Mu), raw Fructus Gardeniae (Zhizi), Radix Paeoniae Rubra (Chi Shao), Cortex Moutan (Mu Dan Pi), Radix Scrophulariae (Xuansheng), Fructus Forsythiae (Lianqiao), Herba Lophatheri (Danzhu), and Radix Glycyrrhizae (Gancao), with  $OB \geq 30\%$  [18] and  $DL \geq 0.18$  [19] for active ingredient screening. Among them, the active ingredients were obtained from Herba Lophatheri and Bubali Cornu through the Chinese Knowledge Infrastructure Project (<http://www.cnki.net/>) and DrugBank (<https://www.drugbank.ca/>) databases. Gypsum fibrosum did not obtain eligible active chemical components in the above databases and can be discarded in this study [20].

### 2.2. Collection of Targets of QBD and COVID-19-related Gene

The TCMSP (<http://tcmspw.com/tcmsp.php>), Swiss Target Prediction (<http://www.swisstargetprediction.ch/>), DrugBank (<https://www.drugbank.ca/>), and PubChem databases (<https://pubchem.ncbi.nlm.nih.gov/>) were used to query the screened targets and PubChem CIDs of the chemical components of QBD. The protein targets were converted into corresponding canonical Gene symbols in the UniProt database (<https://www.uniprot.org/>), and potential targets associated with COVID-19-related genes were screened in the OMIM database (<https://omim.org>).

### 2.3. Interaction Network Construction and Analysis

The potential targets for pneumonia were obtained by taking the intersection of the screened targets of QBD and the targets of pneumonia, and the total interaction network diagram of "ingredient-target-disease" was visualized and screened at the first level using Cytoscape 3.8.2. In the interaction network, different nodes represent Chinese medicine, chemical components, targets and diseases, and the edges between two nodes represent the interactions between nodes. The analysis function of Cytoscape 3.8.2 was used to analyze the topological parameters of the first-level interaction map [21], and the important chemical components were filtered by degree size according to the results of topological parameter analysis.

### 2.4. PPI Network Construction and Core Target Screening Results

The direct targets of anti-pneumonia of QBD were imported into the STRING database for protein-protein interaction network analysis. By setting "Homo sapiens", "minimum required interaction score > 0.9", and "hide disconnected nodes in the network", we obtained the PPI network map [22] and used Cytoscape 3.8.2 to visualize the PPI results and obtain the protein-protein interaction network map [23].

## 2.5. GO Enrichment and KEGG Signaling Pathways Analyses

GO biological process annotation and KEGG pathway enrichment analyses were performed using metaspape (<http://www.metaspape.org>) and microbiology letter (<http://www.bioinformatics.com.cn/>) for the direct action targets of QBD against pneumonia, and the pathways with  $P < 0.01$  were screened [22].

## 2.6. Validation of Key Targets by Molecular Docking

The chemical components and core targets with high degree ranking were screened by the above steps and further validated using molecular docking techniques. The CAS numbers of drug small molecules were obtained from the PubChem database, and the corresponding chemical structures were downloaded from the ZINC database (<http://zinc.docking.org>). The downloaded chemical structures were modified by hydrogenation and charge addition using Pymol and Autodock tools to prepare for molecular docking. The 3D structure of the target protein in the RCSB-PDB database (<http://www.pdb.org>) was downloaded, and the water molecules of the target protein using Pymol were pre-treated [24, 25]. Autodock Tool 1.5.6 was used to explore the possibility of docking between drug small molecules and active sites of target proteins, and the docking structure with the best binding ability was selected for visual analysis and mapping by Pymol.

## 2.7. Animal Experiment Validation

### 2.7.1. Animal Administration and Establishment of Pneumonia Model

Forty male C57BL/6J mice (20-22g) of SPF grade [CertificateNo.: SCXK (Beijing) 2019-0010], were provided by SPF (Beijing) Biotechnology Co., Ltd. Before the start of the experiment, the animals were adaptively fed under standardized feeding conditions for 5 days, with free access to water and food. The temperature in that feed room was 24.5-25.5 degrees C, and the humidity was 45-55%. At the same time, the alternation period of light and shade was 12 h. All experimental animals were approved by the Ethics Committee of the Fifth Medical Center of the General Hospital of the People's Liberation Army (approval No.: IACUC 2020-0049).

After five days of acclimation, the mice were randomly divided into six groups: (1) control group ( $n = 6$ ), (2) LPS pneumonia model group (5 mg/kg,  $n = 6$ ), (3) The dexamethasone group (DEX, 5 mg/kg,  $n = 6$ ), (4) The low dose group of QBD (QBDL, 14.105 g/kg) ( $n = 6$ ), (5) The middle dose group of QBD (QBDM, 28.21 g/kg) ( $n = 6$ ), and (6) High dose of QBD group (QBDH, 56.42 g/kg) ( $n = 6$ ). The control and model groups were administered 0.2ml/10g (dosing volume/body weight) of sterile saline. After 5 days of continuous administration, acute pneumonia was induced by an intratracheal drip of LPS (5 mg/kg) 1 h after the last dose, except in the control group. The control group was dripped with an equal amount of sterile saline. Then, 24h after LPS, molds were taken, and lung tissues were collected for further histological analysis.

### 2.7.2. Histopathological Evaluation and Immunohistochemical Analyses

Mouse left lung lobar tissue was fixed with 4% paraformaldehyde fixative for 48h, dehydrated by ethanol gradient, xylene gradient transparent, routinely paraffin-embedded, sectioned, and hematoxylin-eosin (HE) stained, and lung histopathological structural changes were observed under a light microscope. Immunohistochemistry was performed to detect STAT3, c-JUN, and TNF- $\alpha$  levels. Briefly, paraffin sections of lung tissues were taken, dewaxed in xylene and hydrated in ethanol, incubated dropwise with corresponding primary antibody at 4°C for 90 min, secondary antibody (HRP-labeled) at room temperature for 50 min, developed in DAB chromogenic solution, and hematoxylin solution was used to re-stain cell nuclei, and the sections were sealed after dehydration in gradient ethanol, observed under microscope, and analyzed by Image J software.

## 3. RESULTS

### 3.1. Screening of the Active Ingredients of QBD

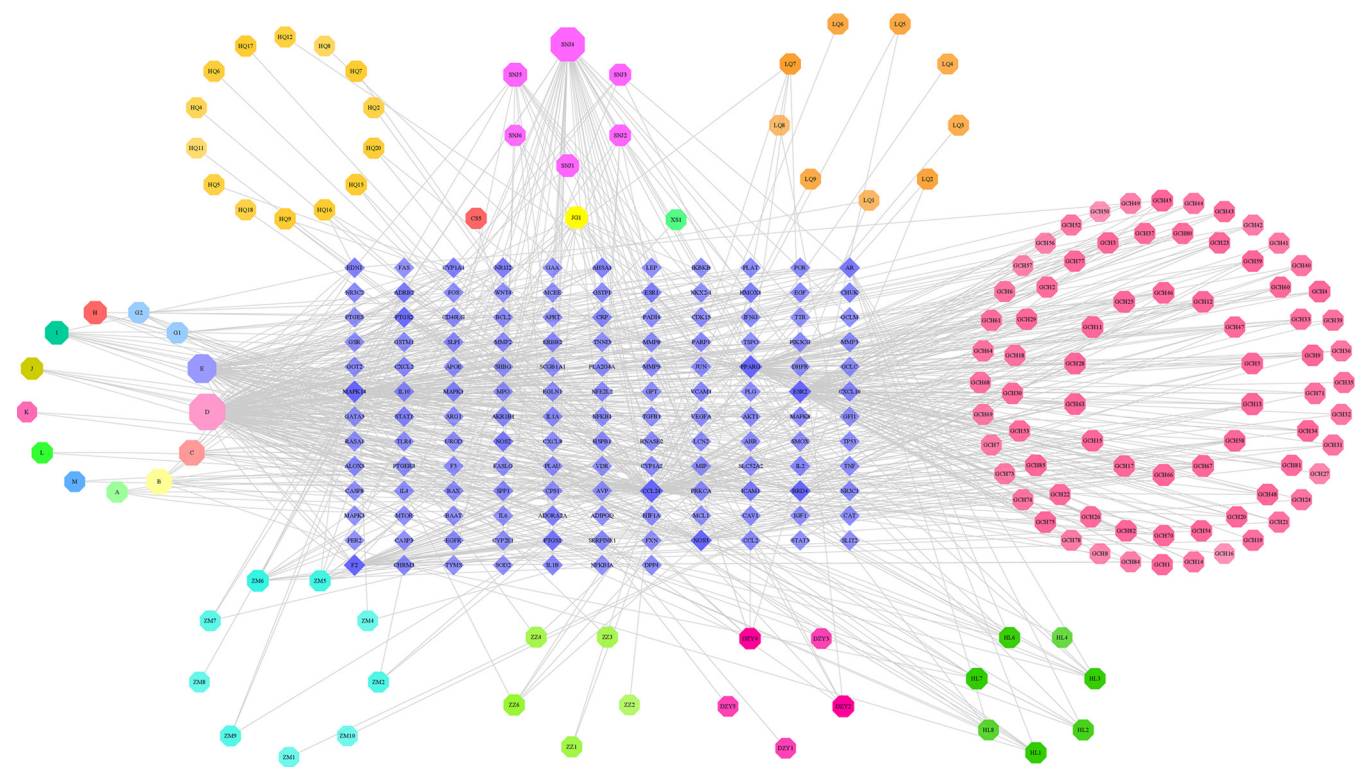
Through TCMSP screening, 176 potential compounds ( $OB \geq 30\%$ ,  $DL \geq 0.18$ ) were identified in the Qingwen Baidu decoction. Meanwhile, 5 active compounds were incorporated from the Chinese Knowledge Infrastructure Project. A total of 181 active compounds were screened from Qingwen Baidu decoction, mainly from Radix Rehmanniae (2 compounds), Bubali Cornu (6 compounds), Radix Scutellariae (32 compounds), Rhizoma Coptidis (11 compounds), Radix Platycodi (4 compounds), Rhizoma Anemarrhenae (12 compounds), raw Fructus Gardeniae (12 compounds), Radix Paeoniae Rubra (14 compounds), Cortex Moutan (6 compounds), Radix Scrophulariae (5 compounds), Fructus Forsythiae (11 compounds), Herba Lophatheri (5 compounds), and Radix Glycyrrhizae (89 compounds). Gypsum fibrosum was excluded because it is a mineral medicine, and the active ingredient is ionic in structure.

### 3.2. Screening and Network Construction of COVID-19-related Gene

In this study, 929 drug targets and 1032 disease targets were screened, and 139 intersecting targets were obtained, which are also called direct-acting targets (Supplementary Fig. 1). The above-obtained drug targets and disease targets were used to construct the "component-target-disease" intersection network (Supplementary Fig. 2). From Supplementary Fig. 2, the direct action targets of Qingwen Baidu decoction were extracted to construct the first level interaction network of "ingredient- direct action target" (Fig. 1). The darker color means more important.

### 3.3. Results of Core Drug Component Screening

In this study, all active ingredients in the first-level interaction network (Fig. 1) were analyzed for topological parameters (Degree, Betweenness Centrality, and Closeness Centrality), and the median values of the three topological parameters were 17, 0.0006792740187183 and 0.305931034482758. In this study, active ingredients with Degree  $\geq 17$ , Betweenness Centrality  $\geq 0.0006792740187183$ , and Closeness Centrality  $\geq 0.305931034482758$  were selected as



**Fig. (1).** Drug-targeted pharmacological network construction. "Component-target-disease" is the first-level interaction network of QBD. The diamond represents the direct-acting target, and the circle represents the chemical component of the drug. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

**Table 1.** Topological parameter analysis of compounds in the first-order interaction network of "component-target-disease" of QBD.

| Degree | Betweenness Centrality | Closeness Centrality | TCMSP Molid | Compound Name                 |
|--------|------------------------|----------------------|-------------|-------------------------------|
| 458    | 0.279054403            | 0.393681221          | MOL000098   | Quercetin                     |
| 398    | 0.569542235            | 0.413343272          | MOL000042   | Alanine                       |
| 240    | 0.079895477            | 0.367096988          | MOL000422   | Kaempferol                    |
| 155    | 0.053594749            | 0.345590527          | MOL000449   | Stigmasterol                  |
| 90     | 0.008832093            | 0.326272433          | MOL000173   | Wogonin                       |
| 74     | 0.029040231            | 0.337184555          | MOL002714   | Baicalein                     |
| 41     | 0.00802342             | 0.330354483          | MOL003896   | 7-Methoxy-2-methyl isoflavone |
| 36     | 0.01129091             | 0.313365357          | MOL000392   | Formononetin                  |
| 36     | 0.014725104            | 0.333734577          | MOL000354   | Isorhamnetin                  |
| 36     | 0.016888894            | 0.333333333          | MOL004373   | Anhydroicaritin               |
| 33     | 0.030711077            | 0.32704217           | MOL004328   | Naringenin                    |
| 33     | 0.007566047            | 0.324174218          | MOL002565   | Medicarpin                    |

important components, namely quercetin, alanine, kaempferol, stigmasterol, wogonin, baicalein, 7-Methoxy-2-methyl isoflavone, formononetin, isorhamnetin, anhydroicaritin, naringenin, and medicarpin, and the screening results are mentioned in Table 1.

3.4. PPI Network Construction and Core Target Screening Results

The direct-acting targets were imported into the STRING database with the minimum required interaction score of 0.9, and the PPI network graph was obtained by removing the

free targets (Fig. 2a), which contained 110 nodes and 499 interaction relationships. The PPI results were visualized and analyzed by Cytoscape 3.8.2, and the protein-protein interaction network maps were obtained by ranking the degree values (Fig. 2b). However, the important targets with a degree greater than 24 were screened (Supplementary Fig. 3).

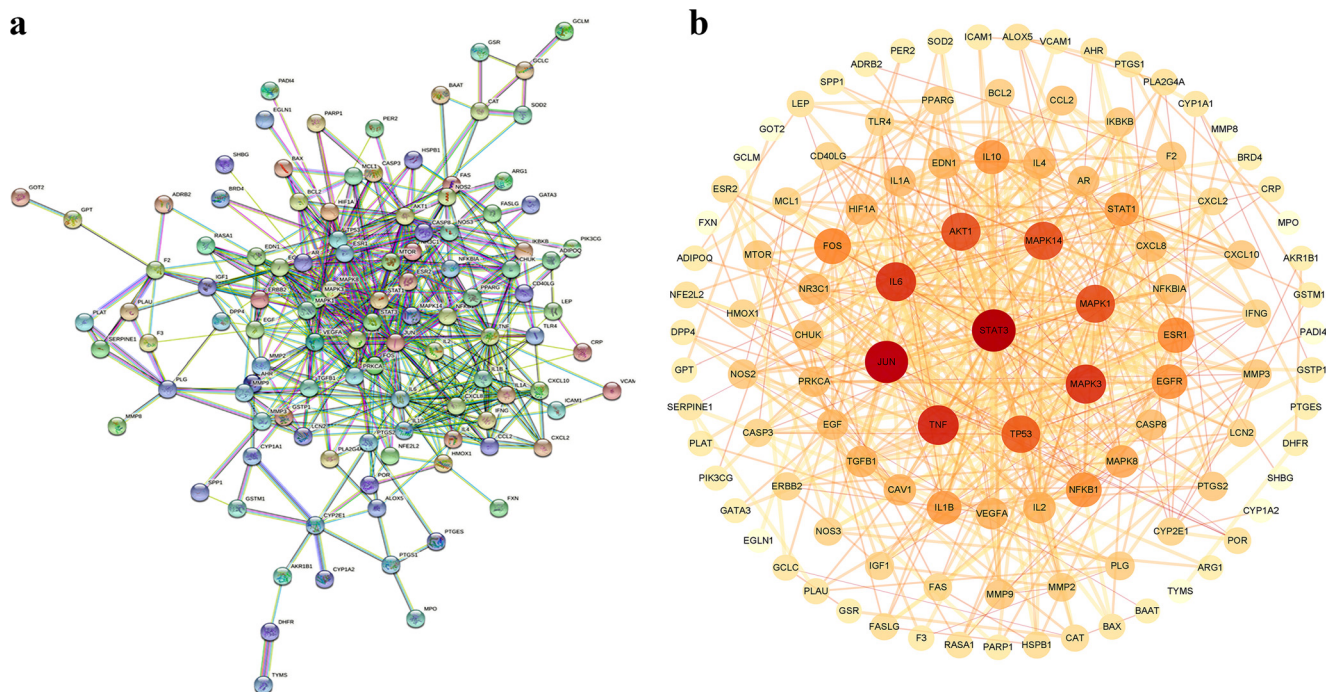
### 3.5. GO and KEGG Enrichment Analyses

The 110 direct-acting target genes obtained from PPI were entered into Metascape (<http://metascape.org/>) for GO enrichment analysis. According to  $p < 0.01$ , 2060 GO entries were obtained related to biological processes, 56 related to cellular components and 124 related to molecular functions. The top 10 entries with high correlation were selected from the results of the three components and summarized into the annotated results of GO enrichment analysis (Supplementary Fig. 4a). The closest biological processes among the gene targets involved in GO were analyzed by the MCODE algorithm in Metascape. The enrichment modules with  $P < 0.01$  and MCODE SCORE  $> 1.25$  were collected, and the core modules of the five biological processes were obtained according to the  $P$  value (Supplementary Fig. 4b). The results showed that the target proteins that interacted tightly in GO(BP) were STAT3, c-JUN, TNF- $\alpha$ , MAPK3, MAPK1, FOS, PPARG, MAPK8, IFNG, NFKB1, etc. The GO biological processes were further analyzed, and the top 20 messages out of 2060 were excerpted (Supplementary Table 1), mostly related to lipopolysaccharide, oxidative stress, positive regulation of cell migration, traumatic stress, peptide response, and apoptosis. The gene targets contained in these 20 entries were visualized with the corresponding biological processes to construct a GO(BP) target protein network diagram (Fig. 3).

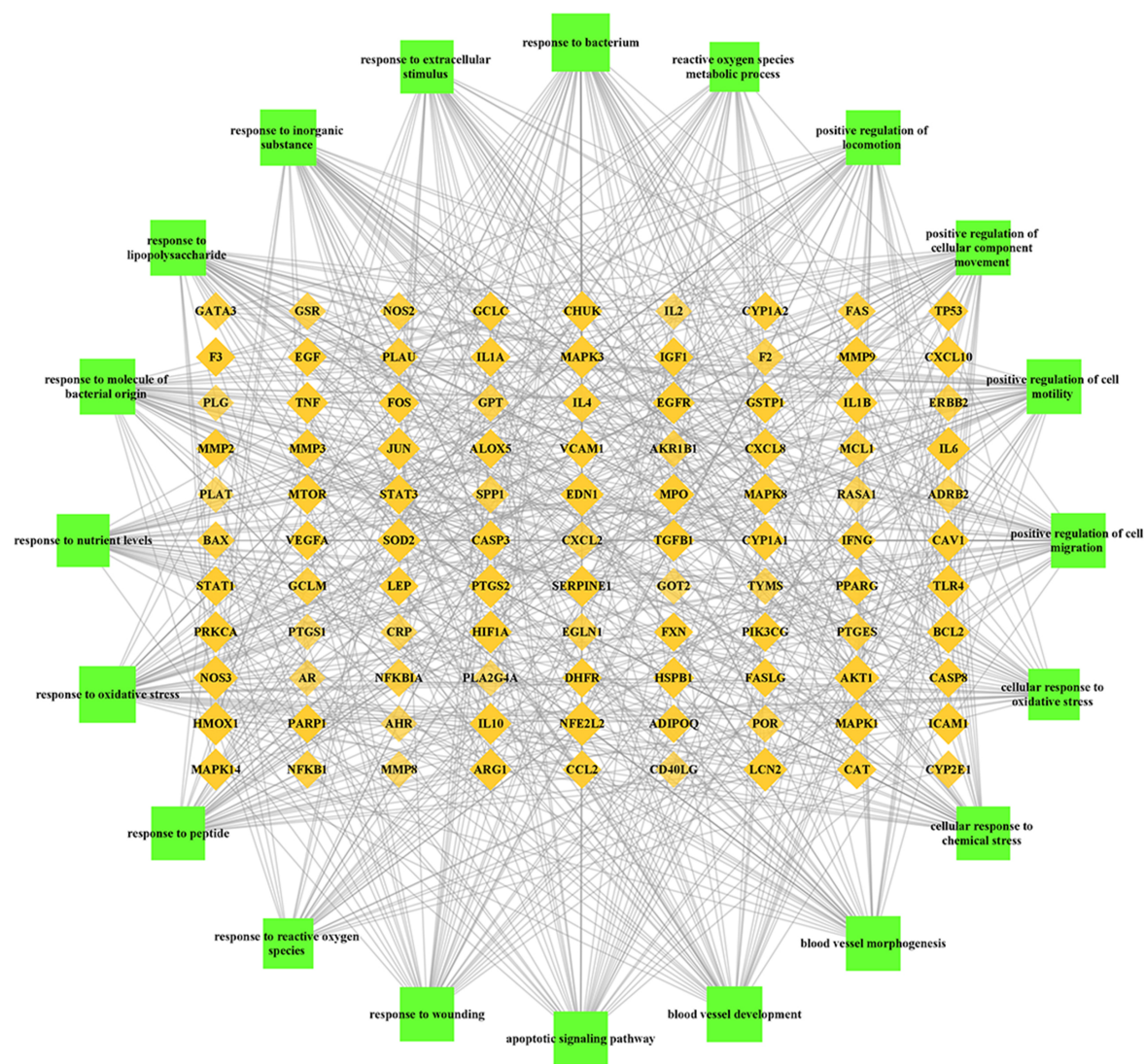
In order to fully reveal the mechanism of action of Qingwen Baidu decoction at the pathway level, KEGG enrichment analysis identified 117 key pathways under the premise of  $p < 0.01$ , indicating that these target genes affect cancer processes, inflammatory pathways, microbial diseases and protein synthesis, etc. The top 20 enriched pathways were screened according to the  $p$ -value (Fig. 4a), mainly involving pathways in cancer, AGE-RAGE signaling pathway in diabetic complications, Chagas disease, HIF-1 signaling pathway, proteoglycans in cancer, non-alcoholic fatty liver disease, inflammatory bowel disease, NF- $\kappa$ B signaling pathway, JAK-STAT signaling pathway, etc. (Fig. 4b). The gene targets contained in these 20 entries were also visualized with the corresponding KEGG to construct a KEGG target protein network diagram (Supplementary Fig. 5), and the KEGG was further analyzed to excise the top 20 messages out of 117 entries (Table 2).

### 3.6. Molecular Docking of Targets and Major Drug Components

In section "3.3", the median values of topological parameters were used to screen the potential chemical components in Qingwen Baidu decoction, and the three chemical components with the highest degree scores, quercetin, alanine, and kaempferol, were used as molecular docking components. The top 3 targets in the "3.4" PPI network diagram, STAT3, c-JUN, and TNF- $\alpha$ , were used as molecular docking target proteins. The above chemical components and targets were molecularly docked, and the conformation with the lowest binding-energy score was selected as the best conformation for analysis during the docking process. The interaction between receptor and ligand was demonstrated in AutoDock



**Fig. (2).** Protein interaction (PPI) network. (a) PPI network exported from the STRING database. (b) Cytoscape visualization of the PPI network. The darker the color, the more important and larger the circle. (A higher resolution / colour version of this figure is available in the electronic copy of the article).



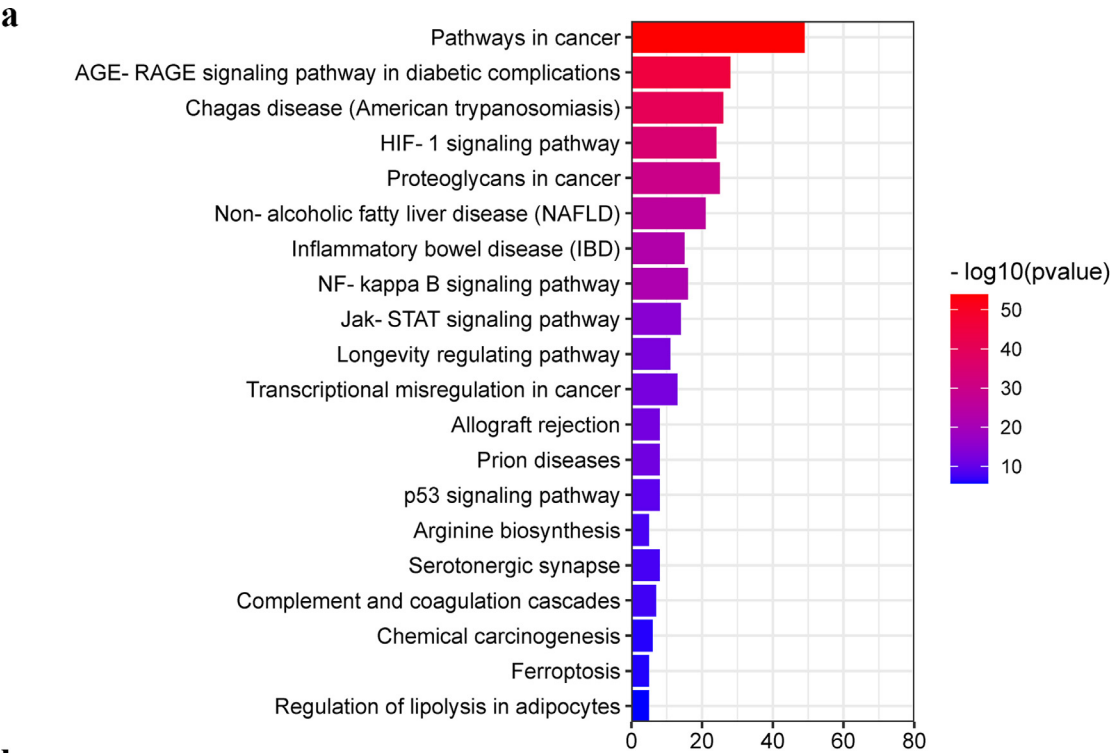
**Fig. (3).** Go (BP) enrichment analysis target protein network diagram. The inner diamond represents the target site, and the outer rectangle represents the GO biological process. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

**Table 2.** Target genes KEGG terms enriched.

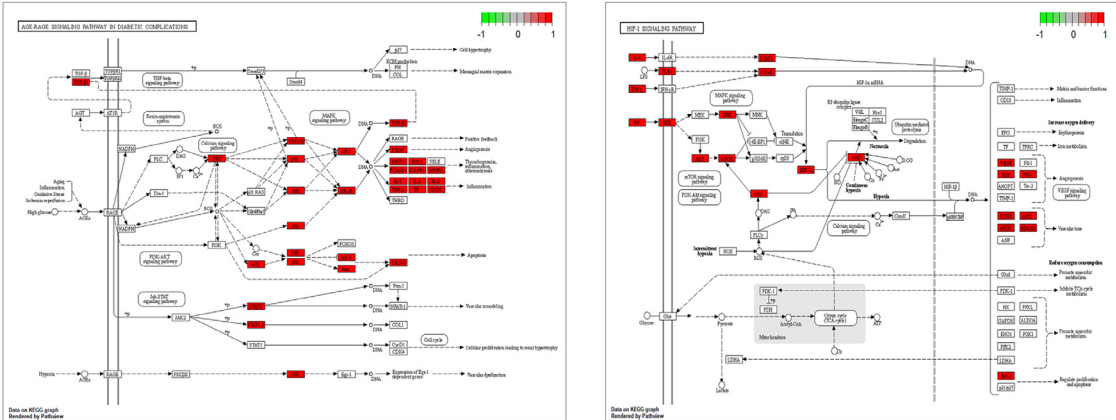
| ID       | Description                                          | Count | Gene Ratio | %     | p-value     | q-value     |
|----------|------------------------------------------------------|-------|------------|-------|-------------|-------------|
| hsa05200 | Pathways in cancer                                   | 49    | 49/110     | 44.55 | 1.17490E-54 | 9.33254E-52 |
| ko04933  | AGE-RAGE signaling pathway in diabetic complications | 28    | 28/110     | 25.45 | 3.54813E-46 | 1.41254E-43 |
| ko05142  | Chagas disease (American trypanosomiasis)            | 26    | 26/110     | 23.64 | 1.58489E-41 | 3.16228E-39 |
| hsa04066 | HIF-1 signaling pathway                              | 24    | 24/110     | 21.82 | 1.28825E-35 | 7.94328E-34 |
| hsa05205 | Proteoglycans in cancer                              | 25    | 25/110     | 22.73 | 1.44544E-30 | 5.01187E-29 |
| hsa04932 | Non-alcoholic fatty liver disease (NAFLD)            | 21    | 21/110     | 19.09 | 1.69824E-26 | 4.16869E-25 |
| ko05321  | Inflammatory bowel disease (IBD)                     | 15    | 15/110     | 13.64 | 1.69824E-23 | 3.01995E-22 |
| ko04064  | NF-kappa B signaling pathway                         | 16    | 16/110     | 14.55 | 1.31826E-22 | 1.99526E-21 |
| hsa04630 | Jak-STAT signaling pathway                           | 14    | 14/110     | 12.73 | 2.29087E-15 | 1.86209E-14 |
| hsa04211 | Longevity regulating pathway                         | 11    | 11/110     | 10    | 2.51189E-13 | 1.62181E-12 |
| hsa05202 | Transcriptional misregulation in cancer              | 13    | 13/110     | 11.82 | 6.02560E-13 | 3.80189E-12 |
| ko05330  | Allograft rejection                                  | 8     | 8/110      | 7.27  | 1.04713E-12 | 6.45654E-12 |

(Table 2) Contd...

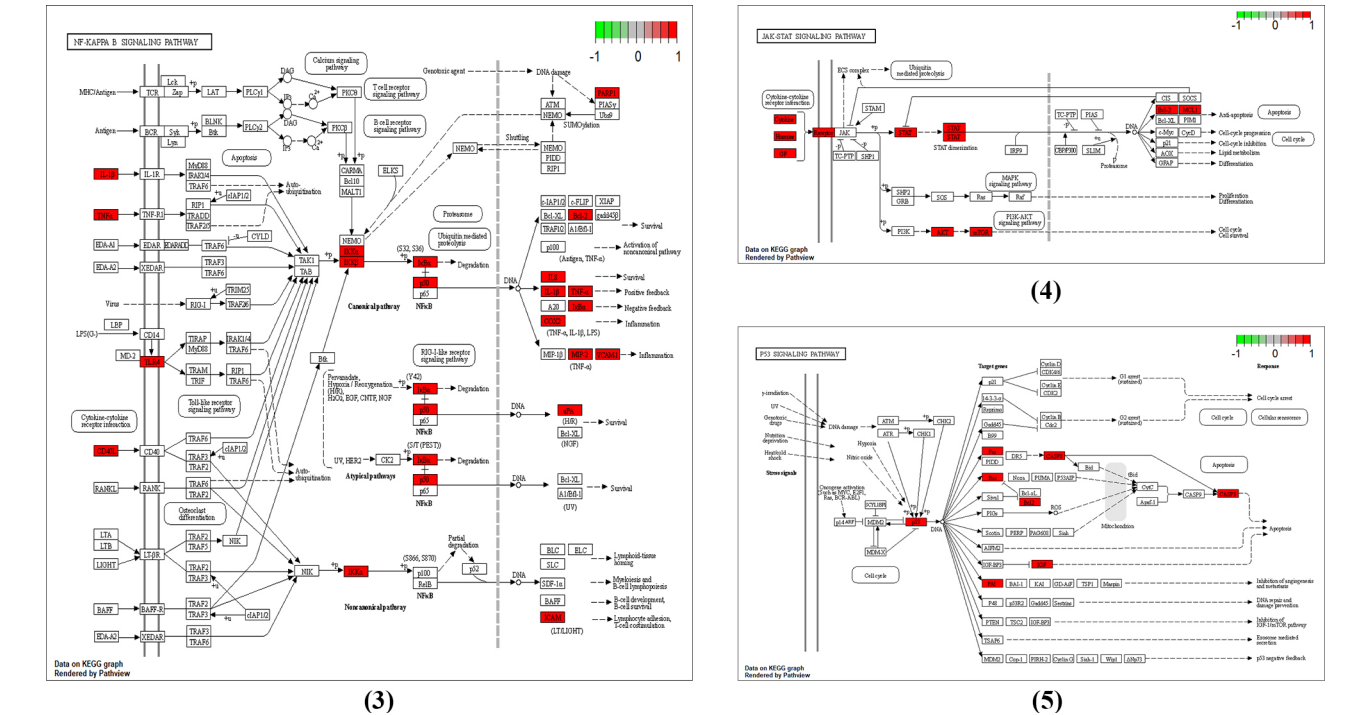
| ID       | Description                           | Count | Gene Ratio | %    | p-value     | q-value     |
|----------|---------------------------------------|-------|------------|------|-------------|-------------|
| hsa05020 | Prion diseases                        | 8     | 8/110      | 7.27 | 3.09030E-12 | 1.77828E-11 |
| hsa04115 | p53 signaling pathway                 | 8     | 8/110      | 7.27 | 5.01187E-10 | 2.39883E-09 |
| ko00220  | Arginine biosynthesis                 | 5     | 5/110      | 4.55 | 1.12202E-08 | 4.89779E-08 |
| hsa04726 | Serotonergic synapse                  | 8     | 8/110      | 7.27 | 1.31826E-08 | 5.75440E-08 |
| hsa04610 | Complement and coagulation cascades   | 7     | 7/110      | 6.36 | 4.36516E-08 | 1.86209E-07 |
| ko05204  | Chemical carcinogenesis               | 6     | 6/110      | 5.45 | 5.62341E-07 | 2.18776E-06 |
| hsa04216 | Ferroptosis                           | 5     | 5/110      | 4.55 | 7.07946E-07 | 2.69153E-06 |
| hsa04923 | Regulation of lipolysis in adipocytes | 5     | 5/110      | 4.55 | 2.29087E-06 | 8.51138E-06 |



**b**



(Fig. 4) Contd...



**Fig. (4).** KEGG enrichment analysis and channel maps. **(a)** KEGG enrichment analysis of target genes. Gene ratio is the ratio of enriched genes to all target genes. **(b)** Pathway maps as 5 important pathways. The red color represents the target genes involved in the pathway. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

**Table 3.** Binding of small drug molecules to core protein targets.

| CAS       | Molecule Name | Molecular Formula                              | MW     | STAT3 Docking Score (kcal/mol) | c-JUN Docking Score (kcal/mol) | TNF-α Docking Score (kcal/mol) |
|-----------|---------------|------------------------------------------------|--------|--------------------------------|--------------------------------|--------------------------------|
| 117-39-5  | Quercetin     | C <sub>15</sub> H <sub>10</sub> O <sub>7</sub> | 302.25 | -3.07                          | -4.51                          | -5.8                           |
| 6898-94-8 | Alanine       | C <sub>3</sub> H <sub>7</sub> NO <sub>2</sub>  | 89.093 | -3.25                          | -4.92                          | -3.05                          |
| 520-18-3  | Kaempferol    | C <sub>15</sub> H <sub>10</sub> O <sub>6</sub> | 286.25 | -3.38                          | -4.52                          | -5.75                          |

Tools Version and Pymol software. From the molecular docking results (Table 3), the binding energies of the three-drug molecules to the three target proteins were less than 1.2 kcal/mol, indicating that both ligands and receptors could bind spontaneously and dock well, and the molecular docking patterns of these nine docking results are displayed in Fig. (5). The high binding affinity of the main active ingredients of Qingwen Baidu decoction to the core targets suggested that Qingwen Baidu decoction may exert its therapeutic effects by modulating the above-related targets.

3.7. Animal Experiment Validation Results

3.7.1. Histopathological Description

In the control group, the alveoli were uniform in size and intact in structure, with normal alveolar epithelial cell morphology, no widening of alveolar intervals, no congestion and bruising of microvessels, and no hemorrhage, edema, or inflammatory cell infiltration in the interstitial spaces of the lungs. In the model group, the alveolar septum was significantly widened, the shape of alveoli was changed, some alveoli were broken or fused, there were exuded erythrocytes in the interstitial space, the interstitial space was

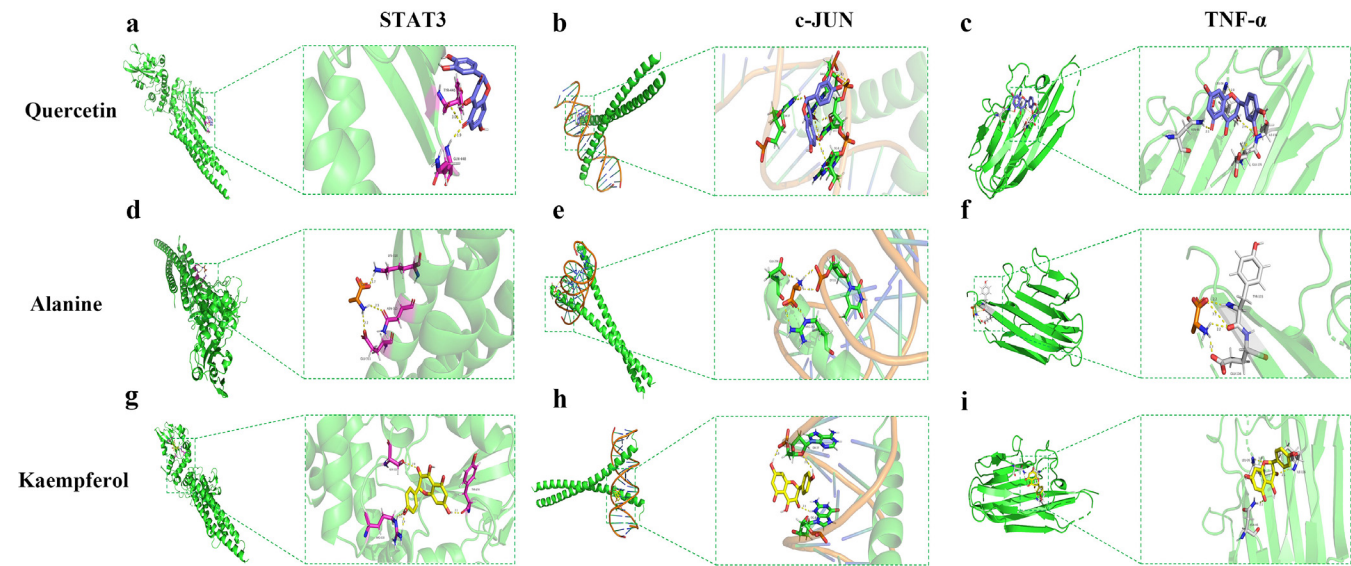
filled with neutrophil-based inflammatory cell infiltration, and the lung tended to be substantial. The dexamethasone group (DEX, 5 mg/kg) reduced the alveolar injury compared to the model group, while in the Qingwen Baidu decoction administration groups (QBDL 14.105 g/kg, QBDM 28.21 g/kg, and QBDH 56.42 g/kg), the pathological injury caused by LPS was found to be significantly improved (Fig. 6).

3.7.2. Immunohistochemical Analysis

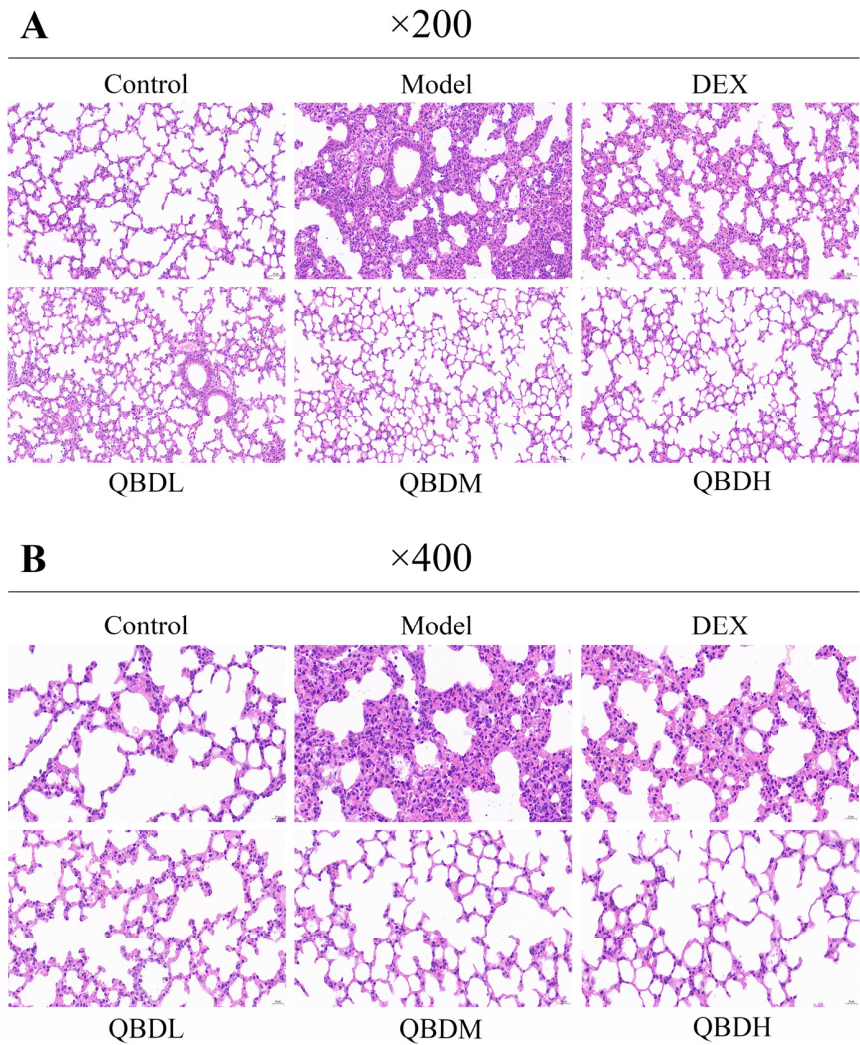
Immunohistochemical analysis showed that STAT3, c-JUN, and TNF-α expressions were significantly higher in the model group than in the normal control group (P<0.01). Interestingly, as shown in Fig. (7), pretreatment with Qingwen Baidu decoction or dexamethasone reversed the above trends compared to the model group, thereby preventing LPS stimulation (P<0.01).

4. DISCUSSION

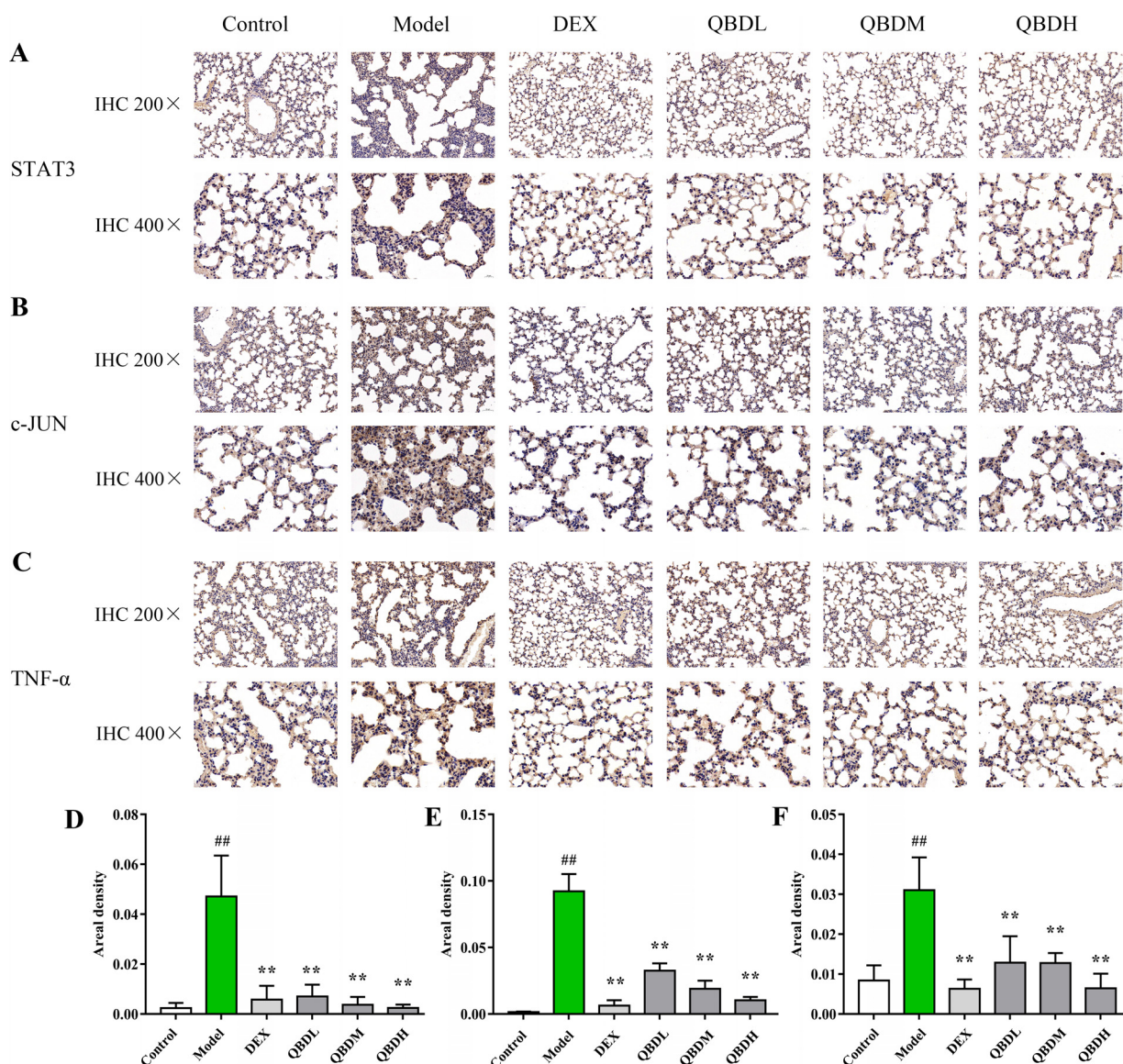
SARS-CoV-2 continues to attack many countries worldwide due to the lack of specific drugs and the mutability of the virus. Notably, China, with its large population, has successfully and effectively controlled the outbreak. Chinese medicine has made outstanding contributions in fighting



**Fig. (5).** (a-i) Diagram of molecular docking pattern. (A higher resolution / colour version of this figure is available in the electronic copy of the article).



**Fig. (6).** (a and b) Effect of Qingwen Baidu decoction on the microscopic appearance of the LPS-induced lung in mice (HE staining,  $\times 200$  and  $\times 400$ ). (A higher resolution / colour version of this figure is available in the electronic copy of the article).



**Fig. (7).** Immunohistochemical methods were used to detect the expression of STAT3, c-JUN, and TNF- $\alpha$  in each group (IHC  $\times 200$  and  $\times 400$ ). (A and D): signal transducer and activator of transcription (STAT3); (B and E): transcription regulatory factor (c-JUN); c,f: tumor necrosis factor (TNF- $\alpha$ ). Factor (c-JUN); (C and F): tumor necrosis factor (TNF- $\alpha$ ). # $P < 0.05$  and ## $P < 0.01$  compared with control group; \* $P < 0.05$  and \*\* $P < 0.01$  compared with model group. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

COVID-19, preventing and controlling SARS-CoV-2 infection, and protecting people's lives and health. In the treatment guideline for combating COVID-19, the "Treatment Protocol for COVID-19" [26], Qingwen Baidu decoction has an indispensable role in the treatment of clinical patients [17] [27]. In this study, we analyzed 181 potentially active chemical components of Qingwen Baidu decoction and established a direct interaction map of 139 target proteins for the treatment of pneumonia. Among these targets, we focused on the top 3 targets, STAT3, c-JUN, and TNF- $\alpha$ , with the highest correlation in the interaction map analysis results and performed molecular docking and animal experiments to verify the interaction between QBD and STAT3, c-JUN, and TNF- $\alpha$  active substances, as well as to deeply demonstrate

the critical therapeutic effects of QBD in pneumonia. The results of this study are the first to demonstrate the effectiveness of QBD in the treatment of COVID-19 from both bioinformatics and animal experiments and provide a blueprint for studying the mechanism of action of QBD. The results of this study may also expand the diversity of clinical drug use in patients with SARS-CoV-2 infection and facilitate targeted drug design and basic research in COVID-19.

We obtained many active ingredients of QBD from the TCMSP database. Rhizoma Coptidis is one of the important components of QBD that acts against the respiratory syncytial virus (RSV) [28] and has a wide range of antibacterial, antiviral, and anti-inflammatory traditional effects. Intri-

guinely, berberine is one of the important extracts of *Rhizoma Coptidis* that acts against dengue virus (DENV) [29], Zika virus (ZIKV) [30], H1N1 influenza virus [31], *Staphylococcus aureus* [32], and may even inhibit coronavirus RNA synthesis as well as virus assembly and release [33]. In addition, berberine can also bind to Nrf2 to exert an antioxidant effect. Therefore, Bousquet [34] proposed that berberine is likely to participate in the interaction of insulin resistance (IR) and novel coronavirus in an Nrf2-dependent or non-dependent way, thereby relieving COVID-19. Palmatine, as another important active substance in *Rhizoma Coptidis*, can not only regulate the levels of glutathione (GSH) and superoxide dismutase (SOD) to achieve antioxidant effects but also reduces the levels of IL-6 and TNF- $\alpha$ , regulates macrophage polarization, and inhibits NF- $\kappa$ B pathway to exert anti-inflammatory activity in the LPS inflammation model, which is consistent with the results of GO and KEGG analysis in this study [35, 36]. These inflammatory mediators and oxidative stress products are considered to be important markers of severity and poor prognosis of COVID-19 disease.

In addition, *Scutellariae Radix* is derived from the root of *Scutellaria baicalensis* Georgi, a plant of Lamiaceae, which is a well-known herbal medicine [37]. It is now officially listed in the Chinese Pharmacopoeia (2020), the European Pharmacopoeia (EP9.0), and the British Pharmacopoeia (BP2018) and is widely used worldwide [38]. Its main active ingredient, baicalin and its aglycone baicalein, reduces lung coefficients and collagen deposition in lung tissue through the TGF- $\beta$ 1-induced ERK1/2 signaling pathway [39], thus reducing substantial lung invasion, which is very helpful for the treatment of patients with severe COVID-19. Surprisingly, baicalein, as a flavonoid, has powerful anti-inflammatory effects like baicalein [40], and both of them have anti-COVID-19 potential.

Quercetin and kaempferol are common components of several herbal medicines in QBD, exerting multi-target and multi-pathway effects, which are consistent with our derived results for the ranking of compound topological parameters and, more importantly, are of interest to many researchers as potential active components against COVID-19. Quercetin is found in *Coptidis Rhizoma*, *Gardeniae Fructus*, *Forsythiae Fructus*, and *Glycyrrhizae Radix et Rhizoma*. It was the first identified inhibitor of the Ebola virus (EBOV) target protein VP24 anti-IFN-I signaling pathway [41]. Mehrbod *et al.* demonstrated that when quercetin derivatives were present at a concentration of 25  $\mu$ g/ml against the H1N1 influenza virus in MDCK cells [42] and even showed comparable resistance to viruses from H1N1 (A/PR/8/34) with strong NA binding ability to zanamivir [43]. Not only that, quercetin also has a strong antiviral therapeutic effect against human respiratory syncytial virus (RSV) [44] and coxsackievirus [45]. Surprisingly, a clinical study [46] showed that quercetin supplementation shortened the time to cure and reduced the severity of symptoms in patients with mild SARS-CoV2. This study provides strong evidence for the anti-COVID-19 potential of quercetin, and many other researchers have presented evidence from different aspects [47-49]. Another flavonoid, kaempferol, which ranks lower than quercetin, has recently attracted much attention as a potential therapeutic target in the pathogenesis of COVID-19 and belongs to the QBDs

*Paeoniae Radix Rubra*, *Gardeniae Fructus*, *Forsythiae Fructus*, *Glycyrrhizae Radix et Rhizoma*, *Moutan Cortex*, and *Anemarrhenae Rhizoma* as the active ingredients. Kaempferol significantly reduced H9N2 virus titers in mice with acute lung injury and exhibited protective effects against H9N2 virus-induced inflammation by inhibiting TLR4 / MyD88-mediated NF- $\kappa$ B and MAPKs pathways [50], which was consistent with the results obtained by our KEGG. In addition, *Rehmanniae Radix* inhibited the replication of hepatitis C virus (HCV) [51], polysaccharides from *Platycodonis Radix* ameliorated respiratory syncytial virus-induced lung injury [52], and ethanolic extract of *Scrophulariae Radix* exerted neuroprotective effects [53]. This shows that a large number of mechanistic studies on these components can predict the efficacy of QBD in preventing SARS-CoV-2 infection, including antiviral, anti-inflammatory, anti-apoptotic, and immune enhancing effects.

KEGG enrichment analysis showed that the key targets were mainly concentrated in the AGE-RAGE signaling pathway, HIF-1 signaling pathway, NF- $\kappa$ B signaling pathway, JAK-STAT signaling pathway, p53 signaling pathway, etc. PPI network analysis showed STAT3, c-JUN, TNF- $\alpha$ , IL-6, and MAPK3 as the core targets, among which the top three ranked STAT3, c-JUN, and TNF- $\alpha$  were selected as HUB genes. Based on these results, we believe that the therapeutic effect of QBD on pneumonia is achieved through the following pathways: (1) AGE-RAGE signaling pathway: Under normal physiological conditions, RAGE is expressed at low levels in the organism, while when the organism is in an inflammatory state, Maillard reaction products form AGEs in a non-enzymatic manner and bind to RAGE to cause abnormally high expression [54]. RAGE is activated, which can further activate the p38 Ras protein and mitogen-activated protein (MAP) kinase pathway to upregulate NF- $\kappa$ B mRNA expression, thereby activating NF- $\kappa$ B. In addition, the AGE-RAGE signaling pathway causes stress in Th17 lymphocytes and regulates IL-17 production [55]. Both the NF- $\kappa$ B pathway and IL-17 production are closely associated with COVID-19, and these results are consistent with the GO analysis involved in the cellular stress response. Interestingly, the AGE-RAGE pathway is often associated with type II diabetes. In a recent study [56], it was found that the mortality rate associated with COVID-19 was significantly higher in diabetic patients than in patients without diabetes. Therefore, it may be beneficial to combat COVID-19 by modulating this pathway in diabetic patients or some elderly groups. Lian *et al.* showed [57] that QBD contains peony bark polysaccharides that can downregulate the production of inflammatory factors (ICAM-1 and VCAM-1) and regulate the TGF- $\beta$ 1 pathway, thereby reducing the serum levels of AGEs and RAGE. (2) JAK-STAT signaling pathway: pro-inflammatory factors and chemokines, such as IL-6, IL-7, and TNF- $\alpha$ , were found in the serum repertoire of critically ill COVID-19 patients and are considered to be a key part of immune stress during SARS-CoV-2 infection, correlating with disease severity and poor clinical outcome [8]. Among them, some cytokines can activate the JAKs-mediated JAK-STAT signaling pathway and perform biological functions, such as inflammatory response and oxidative stress [58]. Researchers started clinical trials to address this phenomenon using the therapeutic effects of JAK inhibitors

in patients with COVID-19. In fact, Cao *et al.* found that patients with severe COVID-19 disease who received ruxolitinib in combination with standard therapy had more significant efficacy [59]. In addition, additional inhibitors of JAK, baricitinib and tofacitinib, are also undergoing larger clinical trials, with baricitinib improving viral replication and abnormal inflammatory responses in 12 patients with moderate COVID-19 in previous studies [60] and tofacitinib aiding in the recovery of one SARS-CoV-2 infected patient [61]. Thus, the JAK-STAT signaling pathway is one of the key inflammatory response cell signaling pathways against COVID-19 [62]. Excitingly, 15 g/kg of QBD was able to improve the physiopathological status of rats with acute lung injury by inhibiting the JAK2/STAT3 signaling pathway and downregulating the protein expression of JAK2, p-JAK2, STAT3 and p-STAT3 in lung tissue [63]. (3) NF- $\kappa$ B signaling pathway: It is well known that the disease severity of COVID-19 is associated with dysregulated immune response and cytokine storm, and NF- $\kappa$ B is one of the key transcription factors of many cytokines [64]. After binding of cytokines to their receptors, the IKK complex activated by signaling, IKK phosphorylates I $\kappa$ B protein and leads to its degradation, further translocating NF- $\kappa$ B to the nucleus and binding to specific DNA binding regions to initiate transcription of downstream genes [65]. Lee *et al.* [66] detected single-cell RNA-seq (scRNA-seq) in eight peripheral blood mononuclear cells (PBMC) from COVID-19 patients and found that the development of severe COVID-19 infection was associated with NF- $\kappa$ B activation. On this basis, a new study demonstrated that the N protein of SARS-CoV-2 can promote the association between TAK1 and IKK complexes, induce NF- $\kappa$ B activation, and thus increase the inflammatory response in COVID-19 patients [67]. Meanwhile, QBD can effectively improve the inflammatory infiltration of lung tissue in rats with acute lung injury by inhibiting the activation of NF- $\kappa$ B and attenuating the inflammatory cascade response, which has a certain lung protective effect [68]. (4) p53 signaling pathway: Previous studies have shown that the tumor suppressor protein p53 is involved in the body's IFN-dependent antiviral activity and can resist the replication of SARS-CoV and human coronavirus NL63 [69]. In addition, p53 and NF- $\kappa$ B are interrelated in the process of inflammation, and p53 ameliorates LPS-induced ALI by inhibiting NF- $\kappa$ B activation [70], which plays a negative role in the development of pneumonia [71]. In computer studies and in silico studies, the S2 subunit of SARS-CoV-2 strongly interacts with p53 [72]. (5) HIF-1 signaling pathway: Hypoxia-inducible factor 1 $\alpha$  (HIF-1 $\alpha$ ) is a key player in the physiological activities of body metabolism, angiogenesis, and cell proliferation [73], especially in glycolysis and the inflammatory response has an indispensable role [74-76]. Inflammation, cytokine storm, and glucose metabolism are closely related to the treatment and prognosis of COVID-19 patients [75-77]. A clinical study performed on whole transcriptome RNA sequencing (RNA-Seq) analysis of blood and bronchoalveolar lavage fluid from 11 COVID-19 patients showed significant upregulation of HIF-1 $\alpha$  and pro-inflammatory cytokine expression [64]. Furthermore, the SARS-CoV-2 ORF3a protein was able to induce the production of mitochondria-derived reactive oxygen species (Mito-ROS) and activate HIF-1 $\alpha$  in SARS-CoV-2-infected human cell lines. Interestingly, HIF-1 $\alpha$  subsequently enhanced viral infection

and pro-inflammatory responses [64]. The researchers employed 2-Deoxy-D-Glucose, a naturally occurring glycolysis inhibitor, to inhibit and mitigate the elevation of HIF-1 $\alpha$  levels in individuals diagnosed with COVID-19 [75, 76, 78].

Interestingly, the AGE-RAGE signaling pathway, JAK-STAT signaling pathway, NF- $\kappa$ B signaling pathway, p53 signaling pathway, and HIF-1 signaling pathway run through the entire course of COVID-19, including cytokine storm, inflammatory response, immune stress, oxidative stress, and glucose metabolism, involved in the development of the disease course and prognosis, consistent with GO biological processes. In further analysis of protein-protein interactions, we focused on STAT3, c-JUN, and TNF- $\alpha$  genes as key nodes of the sub-network and molecular docking between drug small molecules and key proteins. STAT3, a key cytokine in the JAK-STAT pathway, is a member of the STAT family and plays a key role in the inflammatory response [79]. Previous studies have shown that the degree of STAT3 activation in LPS-induced acute lung injury mice was proportional to the degree of inflammatory infiltration of lung tissue and the expression of pro-inflammatory genes (IL-1 $\beta$ , IL-6, TNF- $\alpha$ , *etc.*) [80]. Most importantly, the over-activation of the STAT3/PAI-1 signaling network is usually dependent in patients with severe SARS-CoV-2 infection. Thus, it is reasonable to speculate that STAT3 is a vulnerable point for COVID-19 [81]. Notably, c-JUN, an important transcription factor, is phosphorylated by JNK and is involved in the formation of transcriptional activator protein-1 (AP-1), which regulates the expression of several proteins [82]. In addition, c-JUN is an important player in cell proliferation, differentiation, and apoptosis [83]. Other evidence also suggests that inhibition of c-JUN reduces the transformation of lung epithelial cells in asthma models [84].

In addition, TNF- $\alpha$ , which is mainly produced by macrophages, is one of the most critical factors in the "cytokine storm" and can induce oxidative stress, inflammatory responses [85], and converting enzyme (TACE)-dependent shedding, thereby facilitating viral entry and causing tissue damage through TNF $\alpha$  production [86]. Excess TNF- $\alpha$  has been reported to be found in the plasma and tissues of COVID-19 patients [87]. A recent study found that simultaneous blockade of TNF- $\alpha$  and IFN- $\gamma$  can limit lung tissue injury and inflammation after SARS-CoV-2 infection, which is beneficial for the treatment of COVID-19 patients [88].

Among several common models, such as LPS, smoke, bacterial and chemical stimuli triggered pneumonia, the LPS-induced ALI model is one of the widely used stable and safe experimental models, which resembles many aspects of acute pneumonia disease in humans [89]. Specifically, LPS causes many pathological changes in lung tissue, including hemorrhagic damage, extensive inflammatory cell infiltration, alveolar septal widening, and alveolar lumen fusion. In addition, LPS can cause damage to alveolar capillary membranes, recruitment of neutrophils, activation of oxygen radicals, and inflammatory cytokine formation [90].

These pulmonary histopathological processes are highly compatible with the pulmonary pathological processes in COVID-19 patients [91]. In this study, we used endotracheal drip modeling to focus on the lung and simulate lung infection more realistically. Meanwhile, we identified QBD as an

effective anti-acute pneumonia agent and provided several lines of evidence, including histopathology of lung tissue and immunohistochemical analysis of key target proteins.

## CONCLUSION

In conclusion, network pharmacology showed that the main active components of QBD, including quercetin, alanine, and kaempferol, are able to act on multiple targets. QBD acts on pneumonia mainly through the following pathways: AGE-RAGE signaling pathway, HIF-1 signaling pathway, NF- $\kappa$ B signaling pathway, JAK-STAT signaling pathway, and p53 signaling pathway. Molecular docking results showed that quercetin, alanine, and kaempferol could bind well to the most critical STAT3, c-JUN, and TNF- $\alpha$ , suggesting that they may play an important role in the treatment of pneumonia. Furthermore, our data suggested that QBD treatment ameliorated lung tissue injury in mice with LPS-triggered acute pneumonia. In the future, we may continue to explore the effects of QBD on pneumonia-related genes to further reveal the mechanism of action of anti-COVID-19. In conclusion, this study suggests that QBD may be a candidate drug for the treatment of COVID-19, and the screened pathways and targets may be applied as potential drug targets for the diagnosis or treatment of COVID-19.

## AUTHORS' CONTRIBUTION

Ju Yang, Zhao Zhang, Xiao Ma, and Yanling Zhao conceived the project and wrote the manuscript. Ju Yang and Zhao Zhang performed the main part of the experiments, with contributions from Honghong Liu, Jiawei Wang, and Shuying Xie. Ju Yang, Pengyan Li, Jianxia Wen, and Shizhang Wei contributed to the data collection and analysis. Ruisheng Li provided methods for animal experiments and conditions for animal life. Xiao Ma and Yanling Zhao participated in the project design as well as manuscript draft preparation and revision. All authors read and approved the final manuscript.

## LIST OF ABBREVIATIONS

|          |   |                                               |
|----------|---|-----------------------------------------------|
| CNKI     | = | Chinese Knowledge Infrastructure Project      |
| COVID-19 | = | Coronavirus Disease 2019                      |
| EBOV     | = | Ebola Virus                                   |
| GO       | = | Gene Ontology                                 |
| GSH      | = | Glutathione                                   |
| HE       | = | Hematoxylin-eosin                             |
| IR       | = | Insulin Resistance                            |
| KEGG     | = | Kyoto Encyclopedia of Genes and Genomes       |
| OMIM     | = | Online Mendelian Inheritance in Man databases |
| PPI      | = | Protein-Protein Interaction Network           |
| QBD      | = | Qingwen Baidu Decoction                       |
| SOD      | = | Superoxide Dismutase                          |

|       |   |                                                   |
|-------|---|---------------------------------------------------|
| TCMSP | = | Traditional Chinese Medicine Systems Pharmacology |
| WHO   | = | World Health Organization                         |
| ZIKV  | = | Zika Virus                                        |

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

All experimental animals were approved by the by the Ethics Committee of the Fifth Medical Center of the General Hospital of the People's Liberation Army, China (Approval No.: IACUC 2020-0049).

## HUMAN AND ANIMAL RIGHTS

All animal research procedures were followed in accordance with the standards set forth in the eighth edition of Guide for the Care and Use of Laboratory Animals (published by the National Academy of Sciences, The National Academies Press, Washington, D.C.).

## CONSENT FOR PUBLICATION

Not applicable.

## AVAILABILITY OF DATA AND MATERIALS

All the data and supporting information are provided within the article.

## FUNDING

This work was supported by grants from the National Key R&D Program of China (No. 2018YFC1704504) and China Medical Education Association 2020 major scientific problems and medical technical problems (No.2020K TZ002).

## CONFLICT OF INTEREST

The authors declared no conflict of interest, financial or otherwise.

## ACKNOWLEDGEMENTS

Declared none.

## SUPPLEMENTARY MATERIAL

Supplementary material is available on the publisher's website along with the published article.

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